159102 workshop – Week 2

1. Convert the following binary numbers to decimal. Show all working.

(a) 1011 **(b)** 101010 **(c)** 110011

2. Convert the following decimal numbers to binary. Show all working.

(a) 17 **(b)** 22 **(c)** 35

3. Convert the following binary numbers to hexadecimal. Show all working. You can show that your conversion is correct by converting both the binary and the hexadecimal to decimal.

(a) 1010101 **(b)** 111111 **(c)** 10011101

4. Convert the following hexadecimal numbers to binary. Show all working. You can show that your conversion is correct by converting both the hexadecimal and the binary to decimal.

(a) BE **(b)** 2A **(c)** 1C

5. Perform each of the following subtractions using twos complement arithmetic and a word size of 8 bits. Show all working and check your answers.

(a) 7 - 3 **(b)** 15 - 17 **(c)** 28 - 20

6. The following bit patterns represent values stored in twos complement notation in a computer with a word length of 8 bits. Calculate the decimal value in each case. Show all working.

(a) 0000 1010 **(b)** 1100 1100 **(c)** 0111 1111 **(d)** 1111 0101

7.** The following is a message coded in ASCII using 1 byte per character and then represented in ***hexadecimal notation. Note that 1 byte is 2 hexadecimal digits. What is the message?

6865786164656369616C

***8.** Assume that a computer stores integer numbers in twos complement notation. What are the largest and smallest numbers that can be stored if the computer has a word length of:

(a) four bits **(b)** six bits **(c)** eight bits

***9.** Write a small C++ program that adds integer values in such a way that the result will be greater than maxint. Display the result and show that it is not the expected result.